



LINQ IN C#

Language Integrated Query

Query Syntax · Method Syntax · Operators · Deferred Execution · EF Core · Interview Guide

 **Purpose: Quick Revision Before Interview / Last-Minute Refresh**



Filtering
Where, OfType



Projection
Select, SelectMany



Aggregation
Sum, Count, Avg



Joining
Join, GroupJoin



Grouping
GroupBy, Lookup



Execution
Deferred vs Immediate



EF Core
IQueryable, LINQ-SQL



Interview Q&A
20+ Questions

[Logic Lense](#) | LINQ in C#



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 **How to Use:** Each section builds on the previous. Every operator includes syntax, a real example, and key interview notes. Sections 18–19 are pure interview revision.

What is LINQ?

LINQ (Language Integrated Query) is a set of features introduced in C# 3.0 (.NET 3.5, 2007) that brings native query capabilities directly into the C# language. With LINQ you can query arrays, collections, XML, databases, and more — all using the same consistent syntax with full compile-time type checking and IntelliSense support.

 **One-Liner:** LINQ = Write SQL-like queries in C# that work on ANY data source — arrays, lists, databases, XML — with compile-time safety and IntelliSense.

The Problem LINQ Solves

✗ Without LINQ — Verbose & inconsistent

- Different API for each data source
- No compile-time checking of queries
- Verbose foreach + if + temp lists
- No IntelliSense on string-based queries
- DB queries are raw SQL strings
- XML queries use XPath strings

✓ With LINQ — Concise & uniform

- ONE syntax for all data sources
- Full compile-time type checking
- Concise, readable one-liners
- IntelliSense on every operator
- EF Core translates LINQ to SQL
- LINQ to XML with typed operators

LINQ Data Sources — One Syntax, Many Targets

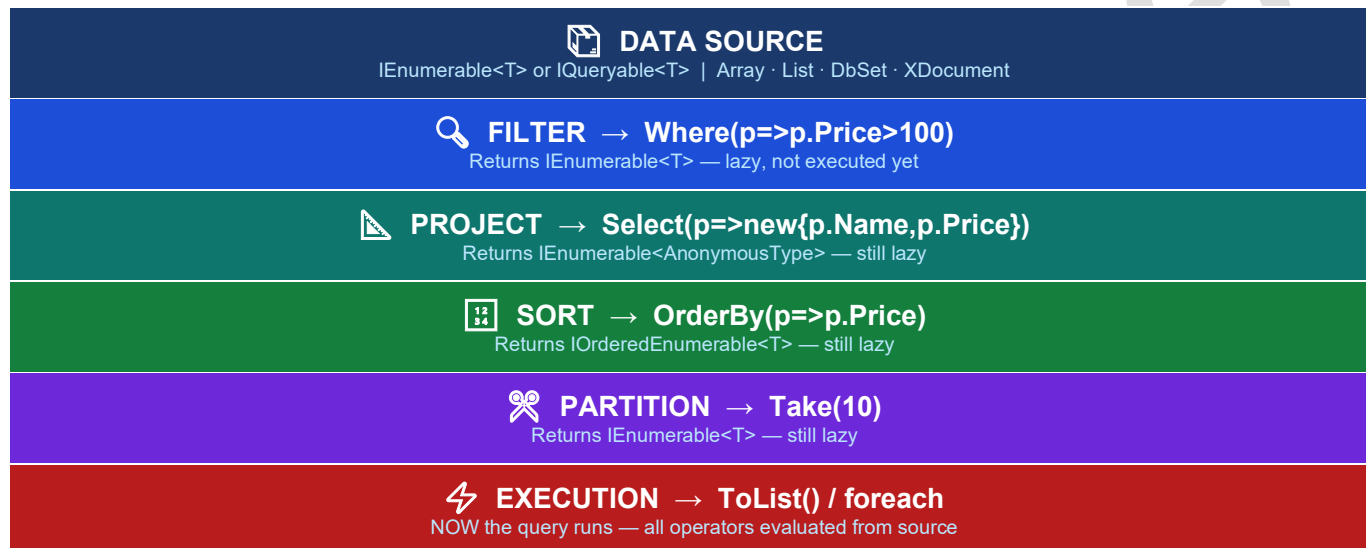
LINQ Variant	Data Source	Namespace / NuGet	Example
LINQ to Objects	Arrays, List<T>, IEnumerable<T>	System.Linq	products.Where(p=>p.Price>50)
LINQ to EF Core	SQL Server, SQLite, PostgreSQL	EF Core	_ctx.Products.Where(p=>p.IsActive)
LINQ to XML	XDocument, XElement	System.Xml.Linq	doc.Descendants("Product")
LINQ to DataSet	ADO.NET DataTable	System.Data	table.AsEnumerable().Where(...)
Parallel LINQ	Multi-core parallel processing	System.Linq (PLINQ)	data.AsParallel().Where(...)
Reactive LINQ	Event streams (Rx.NET)	System.Reactive	observable.Where(x=>x>0)

Key Language Features That Enable LINQ

C# Feature	Purpose in LINQ	Example
Lambda expressions	Concise inline functions passed to operators	<code>p => p.Price > 100</code>
Extension methods	Add operators to <code>IEnumerable<T></code> without changing it	<code>.Where(), .Select(), .OrderBy()</code>
Anonymous types	Project data without creating a class	<code>new { p.Name, p.Price }</code>
Implicitly typed vars	Receive anonymous type results	<code>var result = products.Select(...)</code>
Expression trees	Represent code as data (<code>IQueryable</code>)	Translates LINQ → SQL in EF Core
Object initializers	Create objects in projections	<code>new Product { Name=..., Price=... }</code>
Generics	Type-safe collections and operators	<code>IEnumerable<T>, IQueryable<T></code>

The LINQ Pipeline

A LINQ query consists of three parts: a data source, query operators (the pipeline), and execution. Operators are chained — the output of one becomes the input of the next. The pipeline does not execute until iterated (deferred execution).



 **Key Insight:** Every operator except the final execution step returns an IEnumerable<T> — a RECIPE, not results. The recipe is evaluated only when you iterate it with foreach, ToList(), Count(), First(), etc.

How Extension Methods Power LINQ

LINQ operators are extension methods on IEnumerable<T> defined in System.Linq.Enumerable. They are static methods that look like instance methods due to the "this" keyword on the first parameter.

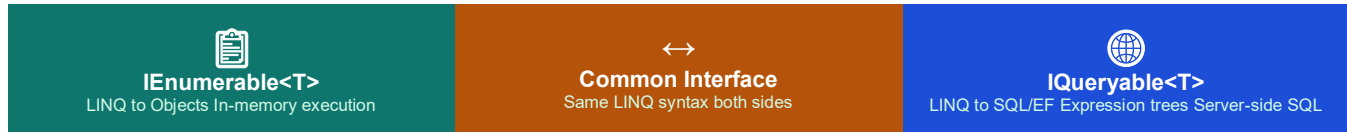
```
// Under the hood — LINQ Where() is an extension method
public static class Enumerable {
    public static IEnumerable<T> Where<T>(
        this IEnumerable<T> source,           // "this" = extension method on IEnumerable

        Func<T, bool> predicate) {
        foreach (var item in source)
            if (predicate(item))             // evaluate predicate per item
                yield return item;           // yield = lazy/deferred
    }
}

// Your code — looks like a method call on the list
var result = products.Where(p => p.Price > 100); // p => ... is the Func<T,bool>
```

```
// Equivalent to: Enumerable.Where(products, p => p.Price > 100)
```

IEnumerable vs IQueryable — The Architecture Split



Expression Trees — How LINQ Becomes SQL

For IQueryable (EF Core), lambdas are compiled to Expression Trees — data structures that represent the code. The EF Core query provider reads the expression tree and generates SQL. This is why the same `.Where(p => p.Price > 100)` becomes `SELECT WHERE Price > 100` on the server.

```
// IEnumerable - lambda compiled to a DELEGATE (executes C# code)
IEnumerable<Product> result1 = products.Where(p => p.Price > 100);
// Runs in C# memory: loads ALL products then filters

// IQueryable - lambda compiled to EXPRESSION TREE (data structure)
IQueryable<Product> result2 = _context.Products.Where(p => p.Price > 100);
// EF Core reads expression tree → generates: SELECT * FROM Products WHERE Price > 100

// SQL filter runs on the DATABASE SERVER - only matching rows returned

// The danger: accidentally switching from IQueryable to IEnumerable
var badQuery = _context.Products           // IQueryable<Product>
    .AsEnumerable()                       // ← forces client-side evaluation!
    .Where(p => p.Price > 100);             // Now runs in C# on ALL products
// SELECT * FROM Products ← loads everything first! Bad for 1M rows
```

Two Syntaxes — Same Result

LINQ has two equivalent syntaxes. Both compile to the same IL (Intermediate Language). Method syntax is more flexible — all operators available. Query syntax is cleaner for complex joins and group-by operations.



QUERY SYNTAX (SQL-like)

```
var result =  
    from p in products  
    where p.Price > 100  
    orderby p.Name  
    select new { p.Name, p.Price };
```



METHOD SYNTAX (Fluent)

```
var result = products  
    .Where(p => p.Price > 100)  
    .OrderBy(p => p.Name)  
    .Select(p => new { p.Name,  
                      p.Price });
```

Comparison Table

Aspect	Query Syntax	Method Syntax
Style	SQL-like keywords (from, where, select)	Chained method calls (.Where(), .Select())
Readability	Cleaner for complex joins, group-by	Cleaner for simple filters and projections
Available operators	Limited subset	ALL operators available
IntelliSense	Good	Excellent — every parameter is typed
Compilation	Compiles to method syntax IL	Directly compiled
Most popular	Less common in modern codebases	Used in 90%+ of modern .NET code
Operator access	No Count(), First(), Aggregate() etc.	Full access to every operator

Side-by-Side Examples

```
// — Setup: sample data —————  
record Product(int Id, string Name, decimal Price, string Category, bool  
IsActive);  
var products = new List<Product> {  
    new(1, "Laptop", 999m, "Electronics", true),  
    new(2, "Phone", 699m, "Electronics", true),  
    new(3, "Desk", 299m, "Furniture", false),  
    new(4, "Chair", 199m, "Furniture", true),  
    new(5, "Monitor", 449m, "Electronics", true),  
};  
  
// — Example 1: Filter + Project —————  
// Query syntax  
var q1 = from p in products where p.IsActive && p.Price > 400 select p.Name;
```

```
// Method syntax
var m1 = products.Where(p => p.IsActive && p.Price > 400).Select(p => p.Name);
// Result: ["Laptop", "Phone", "Monitor"]

// — Example 2: Join —————
record Order(int Id, int ProductId, int Qty);
var orders = new List<Order>{ new(1,1,2), new(2,2,1), new(3,5,3) };

// Query syntax – cleaner for joins
var q2 = from o in orders
         join p in products on o.ProductId equals p.Id
         select new { o.Qty, p.Name, Total = o.Qty * p.Price };

// Method syntax
var m2 = orders.Join(products, o => o.ProductId, p => p.Id,
                    (o, p) => new { o.Qty, p.Name, Total = o.Qty * p.Price });

// — Example 3: Group-by —————
// Query syntax – group-by is cleaner here
var q3 = from p in products group p by p.Category into g
         select new { Category = g.Key, Count = g.Count(), AvgPrice =
g.Average(x=>x.Price) };

// Method syntax
var m3 = products.GroupBy(p => p.Category)
                .Select(g => new { Category = g.Key, Count = g.Count(), AvgPrice =
g.Average(x=>x.Price) });
```

04

Filtering Operators

Where · OfType · Distinct (preview)

Where — The Most Used LINQ Operator

Where filters a sequence, returning only elements that satisfy a predicate. The predicate is a `Func<T, bool>` — a lambda returning true to keep the element. Where is deferred — it builds a query, not a result list.

```
// Basic filter
var expensive = products.Where(p => p.Price > 500);

// Multiple conditions
var activeElectronics = products.Where(p => p.IsActive && p.Category ==
"Electronics");

// Index-based overload — filter with element index
var everyOther = products.Where((p, index) => index % 2 == 0); // keep even-index
items

// Chaining Where — both conditions must be true
var result = products
    .Where(p => p.IsActive)
    .Where(p => p.Price > 200) // equivalent to &&, but more readable
    .Where(p => p.Category != "Furniture");

// Where is lazy — nothing runs yet
IEnumerable<Product> query = products.Where(p => p.Price > 500);
// query describes WHAT to do, not the result
var list = query.ToList(); // NOW it executes and allocates
```

OfType — Filter by Type

OfType<T> filters a non-generic sequence (like an `ArrayList` or `object[]`) returning only elements of the specified type. Safer than `Cast<T>` which throws on wrong type.

```
object[] mixed = { 1, "hello", 2.5, true, 42, "world", 3.14 };

// OfType — keeps only elements that ARE strings
IEnumerable<string> strings = mixed.OfType<string>();
// Result: ["hello", "world"]

IEnumerable<int> ints = mixed.OfType<int>();
// Result: [1, 42]

// vs Cast<T> — THROWS if element is wrong type
// var bad = mixed.Cast<string>().ToList(); // InvalidCastException on 1

// Practical use: inheritance hierarchies
List<Animal> animals = new() { new Dog(), new Cat(), new Dog(), new Fish() };
var dogs = animals.OfType<Dog>().ToList(); // only Dog instances
```

Filtering Performance Tips

Tip	Good	Avoid	Reason
Apply early	<code>.Where()</code> before <code>.Select()</code>	<code>.Select()</code> then <code>.Where()</code>	Filter before projecting reduces data size
Combine conditions	<code>p => p.Active && p.Price > 50</code>	<code>.Where(active).Where(price)</code>	One iteration is faster than two
IQueryable	<code>ctx.Products.Where(predicate)</code>	<code>.ToList().Where(predicate)</code>	DB filter vs in-memory filter on all rows
AsNoTracking	<code>.AsNoTracking().Where(...)</code>	default tracking for reads	Skip EF change-tracker = 30-50% faster reads

Select — Transform Each Element

Select projects each element of a sequence into a new form. It is a one-to-one transformation — for every input element, exactly one output element is produced. Use it to pick specific fields, compute values, or transform types.

```
// Project to a new type
var names      = products.Select(p => p.Name);           //
IEnumerable<string>
var prices     = products.Select(p => p.Price * 1.1m);    // 10% markup
var namePairs  = products.Select(p => (p.Name, p.Price)); // value tuple

// Anonymous type projection — most common pattern
var summaries = products.Select(p => new {
    p.Id,
    p.Name,
    PriceWithTax = p.Price * 1.18m,
    Label = $"{p.Name} ({p.Price:F2})"
});

// Index overload
var indexed = products.Select((p, i) => new { Index = i+1, p.Name });
// Result: { Index=1, Name="Laptop" }, { Index=2, Name="Phone" } ...

// Type conversion — entity to DTO
var dtos = products.Select(p => new ProductDto {
    Id      = p.Id,
    Title   = p.Name.ToUpper(),
    Cost    = p.Price
});
```

SelectMany — Flatten Nested Collections

SelectMany projects each element to a sequence, then FLATTENS all sequences into one. Think of it as "foreach inside foreach". This is the LINQ equivalent of a nested loop with a yield return.

```
// Without SelectMany — nested (not flat)
var nested = orders.Select(o => o.Items); // IEnumerable<IEnumerable<OrderItem>>
// Result: [[item1,item2], [item3], [item4,item5,item6]] — NOT flat

// With SelectMany — flat list of all items
var allItems = orders.SelectMany(o => o.Items); // IEnumerable<OrderItem>
// Result: [item1, item2, item3, item4, item5, item6] — flat!

// SelectMany with result selector — access parent + child together
var orderDetails = orders.SelectMany(
    o      => o.Items,
    (o,i) => new { OrderId = o.Id, Product = i.ProductName, i.Qty }
);

// Common use: split strings
var sentences = new[] { "Hello World", "LINQ is great", "C# rocks" };
var words = sentences.SelectMany(s => s.Split(" "));
// Result: ["Hello","World","LINQ","is","great","C#","rocks"]
```

```
// Flatten nested arrays
int[][] matrix = { new[]{1,2,3}, new[]{4,5}, new[]{6,7,8,9} };
var flat = matrix.SelectMany(row => row);
// Result: [1,2,3,4,5,6,7,8,9]
```

Select vs SelectMany at a Glance

Operator	Input	Output	Use When
Select	T	R (one-to-one)	Transform each item: DTO mapping, computed fields
SelectMany	T	IEnumerable<R> (one-to-many then flat)	Flatten nested collections, split strings
Select + SelectMany	complex	flat	Flatten with access to parent in result selector

Sorting Operators Overview

Operator	Description	Returns
OrderBy(keySelector)	Sort ascending by key	IOrderedEnumerable<T>
OrderByDescending(keySelector)	Sort descending by key	IOrderedEnumerable<T>
ThenBy(keySelector)	Secondary ascending sort	IOrderedEnumerable<T>
ThenByDescending(keySelector)	Secondary descending sort	IOrderedEnumerable<T>
Reverse()	Reverses order of existing sequence	IEnumerable<T>

```
// Single sort - ascending (default)
var byName = products.OrderBy(p => p.Name);
var byPrice = products.OrderBy(p => p.Price);

// Descending
var expensive = products.OrderByDescending(p => p.Price);

// Multi-level sort - MUST use ThenBy (NOT chained OrderBy!)
var sorted = products
    .OrderBy(p => p.Category)           // primary: category A-Z
    .ThenByDescending(p => p.Price)     // secondary: most expensive first
    .ThenBy(p => p.Name);              // tertiary: name A-Z

// ✗ WRONG - chained OrderBy loses the first sort
// products.OrderBy(p => p.Category).OrderBy(p => p.Price) ← only price sort survives!

// Sort strings case-insensitively
var caseSort = products.OrderBy(p => p.Name, StringComparer.OrdinalIgnoreCase);

// Sort with null safety
var nullSafe = products.OrderBy(p => p.Description ?? string.Empty);

// Reverse - reverses existing order (does NOT sort)
var existing = new[] { 5, 1, 3, 2, 4 };
var reversed = existing.Reverse();    // [4, 2, 3, 1, 5] ← not sorted!
var sorted2 = existing.OrderBy(x=>x).Reverse(); // [5,4,3,2,1] ← descending sort
```

⚠ Common Mistake: Never chain multiple OrderBy() calls — only the LAST one survives! Use ThenBy() for secondary sorts. products.OrderBy(x=>x.A).OrderBy(x=>x.B) sorts by B only.

GroupBy — Group Elements by Key

GroupBy groups elements sharing the same key into `IGrouping<TKey, TElement>` objects. It returns `IEnumerable<IGrouping<K,V>>` — a sequence of groups. Each group has a Key and implements `IEnumerable<TElement>`.

```
// Basic group-by
var byCategory = products.GroupBy(p => p.Category);
foreach (var group in byCategory) {
    Console.WriteLine($"Category: {group.Key} ({group.Count()} products)");
    foreach (var product in group)
        Console.WriteLine($"  - {product.Name}: ${product.Price}");
}

// Group then project - most common pattern
var categorySummary = products
    .GroupBy(p => p.Category)
    .Select(g => new {
        Category = g.Key,
        Count    = g.Count(),
        MinPrice = g.Min(p => p.Price),
        MaxPrice = g.Max(p => p.Price),
        Avg      = g.Average(p => p.Price)
    });

// Group by multiple keys - use anonymous type
var byBoth = products.GroupBy(p => new { p.Category, p.IsActive })
    .Select(g => new { g.Key.Category, g.Key.IsActive, Count = g.Count() });

// Element selector - control what is in each group
var namesByCategory = products.GroupBy(
    p      => p.Category,           // key selector
    p      => p.Name                // element selector - only names in each group
);
// Each group.Key = category string, group items = product names (strings)
```

ToLookup vs GroupBy

ToLookup is the immediate (non-deferred) cousin of GroupBy. It builds a dictionary-like structure instantly. Unlike GroupBy (lazy), ToLookup evaluates immediately and never throws — accessing a missing key returns an empty sequence.

```
// ToLookup - evaluates IMMEDIATELY (eager)
ILookup<string, Product> lookup = products.ToLookup(p => p.Category);

// Access like a dictionary - returns IEnumerable<Product> (never null)
var electronics = lookup["Electronics"]; // IEnumerable<Product>
var missing     = lookup["XYZ"];          // empty IEnumerable - no throw!

// GroupBy - DEFERRED (lazy)
var groups = products.GroupBy(p => p.Category);
// groups has not evaluated yet - no data read from source
// Each time you iterate groups, it re-reads the source!

// ToLookup read once, access many times
```

```
var categoryLookup = products.ToLookup(p => p.Category, p => p.Name);
var names1 = categoryLookup["Electronics"]; // [Laptop, Phone, Monitor]
var names2 = categoryLookup["Furniture"];    // [Chair] (Desk is inactive)
```

Feature	GroupBy	ToLookup
Execution	Deferred (lazy)	Immediate (eager)
Returns	IEnumerable<IGrouping<K,V>>	ILookup<K,V>
Missing key	N/A (iterate to find)	Returns empty sequence, no throw
Re-evaluation	Yes — re-reads source	No — built once, reused
Best for	Query pipeline middle	Lookup table used multiple times

Join — Inner Join Between Two Sequences

Join performs an inner join between two sequences based on matching keys. Only elements with matching keys in BOTH sequences appear in the result — like SQL INNER JOIN.

```
// Join signature: source.Join(inner, outerKey, innerKey, resultSelector)
var orderDetails = orders.Join(
    products,                // inner sequence
    o => o.ProductId,        // outer key selector
    p => p.Id,                // inner key selector
    (o, p) => new {           // result projection
        OrderId = o.Id,
        Product  = p.Name,
        UnitPrice = p.Price,
        Qty      = o.Qty,
        Total    = o.Qty * p.Price
    }
);

// Query syntax — cleaner for joins
var q = from o in orders
        join p in products on o.ProductId equals p.Id
        select new { o.Id, p.Name, Total = o.Qty * p.Price };

// Multiple joins
var full = from o in orders
           join p in products on o.ProductId equals p.Id
           join c in customers on o.CustomerId equals c.Id
           select new { o.Id, ProductName = p.Name, CustomerName = c.Name };
```

GroupJoin — Left Outer Join

GroupJoin performs a left outer join. For each element in the outer (left) sequence, it collects ALL matching elements from the inner (right) sequence into a group — even if there are zero matches. This is SQL LEFT OUTER JOIN.

```
// GroupJoin = left outer join
var customersWithOrders = customers.GroupJoin(
    orders,
    c => c.Id,
    o => o.CustomerId,
    (customer, orderGroup) => new {
        Customer    = customer.Name,
        OrderCount  = orderGroup.Count(),
        Orders      = orderGroup
    }
);

// Includes customers with ZERO orders — unlike inner Join

// Flatten to mimic SQL LEFT JOIN with DefaultIfEmpty
var leftJoin = customers
    .GroupJoin(orders, c=>c.Id, o=>o.CustomerId, (c,og) => new{c,og})
    .SelectMany(x => x.og.DefaultIfEmpty(), // null if no orders
        (x, o) => new { CustomerName = x.c.Name, OrderId = o?.Id });
```

Zip — Pair Elements from Two Sequences

Zip merges two sequences element-by-element into tuples or custom results. If sequences have different lengths, it stops at the shorter one.

```
var names = new[] { "Alice", "Bob", "Carol" };
var scores = new[] { 95, 87, 91 };

// Zip into tuples (C# 7+)
var paired = names.Zip(scores);
// [(Alice,95), (Bob,87), (Carol,91)]

// Zip with result selector
var labeled = names.Zip(scores, (name, score) => $"{name}: {score}");
// ["Alice: 95", "Bob: 87", "Carol: 91"]

// Three-way Zip (.NET 6+)
var ranks = new[] { "Gold", "Silver", "Bronze" };
var result = names.Zip(scores).Zip(ranks,
    (ns, rank) => $"{rank}: {ns.First} ({ns.Second})");
```

Built-in Aggregation Operators

```
var prices = products.Select(p => p.Price).ToList();
// [999, 699, 299, 199, 449]

// Count — number of elements
int total      = products.Count();           // 5
int active     = products.Count(p => p.IsActive); // 4

// Sum
decimal sum    = products.Sum(p => p.Price); // 2645

// Min / Max
decimal min    = products.Min(p => p.Price); // 199
decimal max    = products.Max(p => p.Price); // 999
var cheapest   = products.MinBy(p => p.Price); // Product with Price=199 (.NET 6+)
var priciest   = products.MaxBy(p => p.Price); // Product with Price=999 (.NET 6+)

// Average
double avg     = products.Average(p => (double)p.Price); // 529

// LongCount — for collections > int.MaxValue
long bigCount  = bigCollection.LongCount();

// Aggregate — general-purpose fold/reduce
decimal product = new[] { 1m, 2m, 3m, 4m, 5m }.Aggregate((acc, n) => acc * n); // 120

// Aggregate with seed
string sentence = new[] { "Hello", "world", "from", "LINQ" }
    .Aggregate("Start:", (acc, word) => acc + " " + word);
// "Start: Hello world from LINQ"

// Aggregate with seed + result selector
string csv = products.Aggregate(
    new System.Text.StringBuilder(),
    (sb, p) => sb.Append(p.Name).Append(","),
    sb => sb.ToString().TrimEnd(",").ToCharArray());
// "Laptop,Phone,Desk,Chair,Monitor"
```

Aggregate Operator Reference

Operator	Returns	Throws if Empty	With Predicate
Count()	int	No — returns 0	Count(predicate) — count matching
LongCount()	long	No — returns 0	LongCount(predicate)
Sum(selector)	numeric	No — returns 0	Sum(p => p.Price)
Min(selector)	T	Yes — InvalidOp	Min(p => p.Price)
Max(selector)	T	Yes — InvalidOp	Max(p => p.Price)

Operator	Returns	Throws if Empty	With Predicate
Average(selector)	double	Yes — InvalidOp	Average(p => (double)p.Price)
Aggregate(func)	T	Yes — InvalidOp	Aggregate(seed, func) — no throw
MinBy(.NET6)	T?	No — returns null	MinBy(p => p.Price)
MaxBy(.NET6)	T?	No — returns null	MaxBy(p => p.Price)

⚠ Min/Max throw on empty sequence: Always call Any() first or use Min(selector) with a default: `products.Min(p => (decimal?)p.Price) ?? 0m`

Set Operation Operators

```
var a = new[] { 1, 2, 3, 4, 5 };
var b = new[] { 3, 4, 5, 6, 7 };

// Distinct - remove duplicates
var nums = new[] { 1, 2, 2, 3, 3, 3, 4 };
var unique = nums.Distinct(); // [1, 2, 3, 4]

// DistinctBy (.NET 6+) - deduplicate by key
var uniqueByCategory = products.DistinctBy(p => p.Category);
// One product per category (first encountered)

// Union - all unique elements from BOTH
var union = a.Union(b); // [1,2,3,4,5,6,7]

// Intersect - only elements in BOTH
var intersect = a.Intersect(b); // [3,4,5]

// Except - elements in A but NOT in B
var except = a.Except(b); // [1,2]

// Set operations with custom equality (override GetHashCode+Equals OR use comparer)
var distinctProducts = products.Distinct(new ProductEqualityComparer());

// IntersectBy / ExceptBy / UnionBy (.NET 6+) - set ops by key
var activeNames = activeProducts.Select(p => p.Name);
var cheapNames = cheapProducts.Select(p => p.Name);
var activeCheap = products.IntersectBy(cheapNames, p => p.Name);

// Practical use: find new/removed items between two lists
var added = newList.Except(oldList); // items in new but not old
var removed = oldList.Except(newList); // items in old but not new
var same = newList.Intersect(oldList); // items in both
```

Set Operations Visual Reference

Operation	Visual	Result from A=[1,2,3,4,5] B=[3,4,5,6,7]	SQL Equivalent
Distinct()	Remove dupes from A	[1,2,3,4,5] (unchanged if already unique)	SELECT DISTINCT
Union(B)	A + B, no duplicates	[1,2,3,4,5,6,7]	UNION
Intersect(B)	Common elements	[3,4,5]	INTERSECT
Except(B)	In A but not in B	[1,2]	EXCEPT / MINUS

Partitioning Operators

```
var nums = new[] { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };

// Take - first N elements
var first3 = nums.Take(3); // [1,2,3]

// Skip - skip first N elements
var after3 = nums.Skip(3); // [4,5,6,7,8,9,10]

// Combined for pagination
int page = 2, size = 3;
var page2 = nums.Skip((page-1)*size).Take(size); // [4,5,6]

// Take(Range) .NET 8+
var middle = nums.Take(3..6); // [4,5,6] (index 3 to 5 inclusive)
var last3 = nums.Take(^3..); // [8,9,10]

// TakeWhile - take while predicate is true (stops at first false)
var smallOnes = nums.TakeWhile(n => n < 5); // [1,2,3,4] stops at 5

// SkipWhile - skip while predicate is true (starts at first false)
var afterSmall = nums.SkipWhile(n => n < 5); // [5,6,7,8,9,10] starts at 5

// Chunk (.NET 6+) - split into chunks of N
var chunks = nums.Chunk(3);
// [[1,2,3], [4,5,6], [7,8,9], [10]] last chunk may be smaller

// Practical: batch processing
foreach (var batch in largeList.Chunk(100))
    await ProcessBatchAsync(batch);
```

Pagination Pattern with LINQ

```
// Standard pagination helper
public static IEnumerable<T> Paginate<T>(
    IEnumerable<T> source, int pageNumber, int pageSize)
    => source.Skip((pageNumber - 1) * pageSize).Take(pageSize);

// EF Core - server-side pagination (ONLY take matched rows from DB)
public async Task<PagedResult<ProductDto>> GetPagedAsync(
    int page, int size, CancellationToken ct)
{
    var query = _ctx.Products.Where(p => p.IsActive).AsNoTracking();
    var total = await query.CountAsync(ct); // one COUNT query
    var items = await query
        .OrderBy(p => p.Name)
        .Skip((page-1)*size)
        .Take(size)
        .Select(p => new ProductDto(p.Id, p.Name, p.Price))
        .ToListAsync(ct); // SQL: SELECT ... OFFSET
    ... FETCH
    return new PagedResult<ProductDto>(items, total, page, size);
}
```

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Element Operators

First · Single · Last · ElementAt · DefaultIfEmpty

Element Operator Reference

Operator	Returns	If Empty Sequence	If No Match (predicate)
First()	First element	InvalidOperationException	N/A
First(predicate)	First matching	InvalidOperationException	InvalidOperationException
FirstOrDefault()	First or default(T)	default(T) — null/0	N/A
FirstOrDefault(pred)	First matching or def	default(T)	default(T)
Last()	Last element	InvalidOperationException	N/A
LastOrDefault()	Last or default(T)	default(T)	N/A
Single()	The ONE element	InvalidOperationException	N/A
Single(predicate)	The ONE matching	InvalidOperationException	InvalidOperationException
SingleOrDefault()	One or default(T)	default(T)	default(T)
ElementAt(i)	Element at index i	ArgumentOutOfRangeException	N/A
ElementAtOrDefault(i)	Element at i or default	default(T)	N/A
DefaultIfEmpty()	Source or [default]	IEnumerable with one default	N/A

```
// First — grabs the first, throws if empty
var first = products.First();
var laptop = products.First(p => p.Name == "Laptop");

// FirstOrDefault — returns null instead of throwing
var maybe = products.FirstOrDefault(p => p.Name == "Tablet"); // null

// Single — enforces EXACTLY one result
var byId = products.Single(p => p.Id == 3); // throws if 0 or 2+ found

// SingleOrDefault — null if not found, throws if more than one
var maybe2 = products.SingleOrDefault(p => p.Id == 999); // null

// When to use Single vs First:
// Single → you EXPECT exactly one (e.g., lookup by unique ID)
// First → you want the best match and there might be many

// DefaultIfEmpty — used for left join patterns
```

```
var orDefault = products
    .Where(p => p.Price > 10000) // no results
    .DefaultIfEmpty(new Product(0, "None", 0m, "", false))
    .First(); // returns the default product, not throw

// C# 7.1+ - specify default inline
var first2 = products.FirstOrDefault(p => p.Id==999) ?? new
Product(0, "", 0m, "", false);
```

💡 **Single vs First:** Use `Single()` when business logic says exactly one match must exist (e.g., find user by email — should be unique). Use `First()` when you want the top result from many. `Single()` scans the whole sequence to ensure no second match exists — $O(n)$.

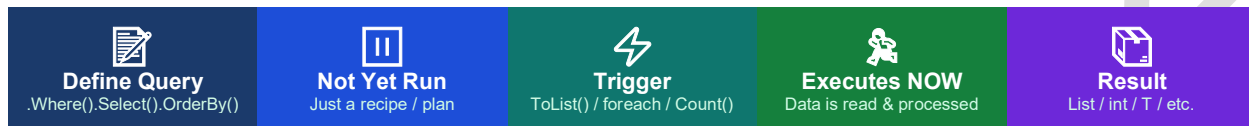
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Deferred vs Immediate Execution

Lazy · When Queries Run · Snapshot vs Live · Pitfalls

The Golden Rule of LINQ Execution

Most LINQ operators are LAZY — they build a description of what to do (a query object), but do NOT execute until the result is actually needed. Execution is triggered by: iteration (foreach), conversion (ToList, ToArray), aggregation (Count, Sum, First), or EF Core awaiting.



```
// DEFERRED — query is a recipe, not data
IEnumerable<string> query = products
    .Where(p => p.Price > 100)    // still lazy
    .Select(p => p.Name);        // still lazy

// At this point: NO iteration has happened!
products.Add(new Product(6, "Tablet", 299m, "Electronics", true)); // add after
query defined

// Trigger execution — NOW the query runs against the CURRENT list
var result = query.ToList(); // includes "Tablet" because it was added before
execution!

// — IMMEDIATE operators (execute on the spot) —
var count    = products.Count(); // runs immediately, returns
int
var list     = products.Where(p=>p.IsActive).ToList(); // runs immediately
var arr      = products.Select(p=>p.Name).ToArray(); // runs immediately
var dict     = products.ToDictionary(p=>p.Id); // runs immediately
var lookup   = products.ToLookup(p=>p.Category); // runs immediately
var first    = products.First(p=>p.Price<200); // runs immediately
var total    = products.Sum(p=>p.Price); // runs immediately

// PITFALL — multiple enumeration
var names = products.Where(p=>p.IsActive).Select(p=>p.Name);
Console.WriteLine(names.Count()); // 1st execution
foreach (var n in names) { } // 2nd execution — reads source AGAIN
// Fix: var names = products.Where(...).Select(...).ToList(); // ONE execution
```

Deferred vs Immediate Reference

Category	Operators	Executes When
Deferred — filtering	Where, OfType	First foreach / ToList() call
Deferred — projection	Select, SelectMany	First foreach / ToList() call
Deferred — sorting	OrderBy, ThenBy, Reverse	First foreach / ToList() call
Deferred — partitioning	Take, Skip, TakeWhile, SkipWhile	First foreach / ToList() call
Deferred — joining	Join, GroupJoin, Zip	First foreach / ToList() call

Category	Operators	Executes When
Deferred — grouping	GroupBy	First foreach / ToList() call
Immediate — conversion	ToList, ToArray, ToDictionary, ToLookup	Immediately at the call site
Immediate — aggregation	Count, Sum, Min, Max, Average, Aggregate	Immediately at the call site
Immediate — element	First, Single, Last, ElementAt	Immediately at the call site
Immediate — quantifier	Any, All, Contains	Immediately at the call site

The Critical Difference

IEnumerable<T>

- LINQ to Objects (in-memory)
- Lambda compiles to DELEGATE
- Executes C# code in memory
- All data loaded first, then filtered
- No translation — what you write runs
- Best for: List, Array, in-memory data

IQueryable<T>

- LINQ to SQL / EF Core (server-side)
- Lambda compiles to EXPRESSION TREE
- Expression tree translated to SQL
- Filter runs on database server
- Only matching rows returned to C#
- Best for: DbSet, database queries

```
// — IEnumerable — in-memory filtering —
IEnumerable<Product> list = GetAllProducts(); // loads ALL products into memory
var result1 = list.Where(p => p.Price > 100); // C# foreach in memory

// — IQueryable — server-side filtering —
IQueryable<Product> dbQuery = _ctx.Products; // NO data loaded yet
var result2 = dbQuery.Where(p => p.Price > 100); // adds WHERE to SQL
var data = await result2.ToListAsync(); // NOW: SELECT * WHERE Price>100
// Only expensive products come back from the DB — huge performance difference!

// — The dangerous AsEnumerable() switch —
var bad = _ctx.Products // IQueryable<Product>
    .AsEnumerable() // ← switch to IEnumerable! ALL rows loaded
    .Where(p => p.Price > 100); // C# filter AFTER loading everything
// SELECT * FROM Products (gets 1,000,000 rows!)

// — When AsEnumerable() is legitimate —
// Use when EF Core cannot translate a C# method to SQL
var result3 = _ctx.Products
    .Where(p => p.IsActive) // server-side: goes to SQL
    .AsEnumerable() // client-side after active filter
    .Where(p => MyComplexMethod(p.Name)); // cannot be translated to SQL
```

IQueryable Method Chain & SQL Generation

```
// Every method adds to the SQL query — nothing runs until ToListAsync()
var query = _ctx.Products // SELECT * FROM Products
    .Where(p => !p.IsDeleted) // WHERE IsDeleted = 0
    .Where(p => p.CategoryId == categoryId) // AND CategoryId = @p0
    .OrderBy(p => p.Price) // ORDER BY Price
    .Skip((page-1)*size) // OFFSET @p1 ROWS
    .Take(size) // FETCH NEXT @p2 ROWS
    .Select(p => new { p.Id, p.Name, p.Price }); // SELECT Id, Name, Price

// Generated SQL (SQL Server):
// SELECT p.Id, p.Name, p.Price
// FROM Products AS p
// WHERE p.IsDeleted = 0 AND p.CategoryId = @p0
// ORDER BY p.Price
// OFFSET @p1 ROWS FETCH NEXT @p2 ROWS ONLY

var results = await query.ToListAsync(); // executes the complete SQL above
```

What is yield return?

yield return transforms a method into a lazy iterator. The method body runs only when the caller moves to the next element. State is preserved between calls — no full collection is built in memory. This is exactly how LINQ operators like Where() work internally.

```
// Normal method - builds full list in memory
IEnumerable<int> EvenNumbers_Eager(int max) {
    var result = new List<int>();
    for (int i=0; i<=max; i++)
        if (i%2==0) result.Add(i);
    return result; // all values computed upfront
}

// yield return method - lazy iterator
IEnumerable<int> EvenNumbers_Lazy(int max) {
    for (int i=0; i<=max; i++)
        if (i%2==0) yield return i; // pause here, give value, resume on next
    call
}

// Infinite sequence - only possible with yield
IEnumerable<long> Fibonacci() {
    long a=0, b=1;
    while (true) {
        yield return a;
        (a, b) = (b, a+b);
    }
}
var first20 = Fibonacci().Take(20).ToList(); // only 20 values computed

// yield break - end the sequence early
IEnumerable<int> TakePositive(IEnumerable<int> src) {
    foreach (var n in src) {
        if (n < 0) yield break; // stop when negative found
        yield return n;
    }
}
```

How LINQ Where() Uses yield Internally

```
// LINQ Where() simplified implementation
public static IEnumerable<T> Where<T>(this IEnumerable<T> source, Func<T,bool>
predicate) {
    foreach (var item in source) {
        if (predicate(item))
            yield return item; // lazy: caller gets ONE item, then we pause
    }
}

// This means chaining Where().Select().Take() is super efficient:
var result = hugeList
    .Where(x => x > 0) // lazily filters - yields one item at a time
    .Select(x => x * 2) // lazily projects - passes item forward
    .Take(5) // stops after 5 - previous operators also stop
    .ToList();
```

```
// ONLY reads enough source items to produce 5 results – not the whole list!
```

Logic Lense

EF Core LINQ Best Practices

```
// — 1. Use AsNoTracking() for read-only queries —
var products = await _ctx.Products
    .AsNoTracking()           // skip change tracker = 30-50% faster
    .Where(p => p.IsActive)
    .ToListAsync();

// — 2. Include for eager loading - avoid N+1 —
// ✗ N+1 problem - 1 query for orders + 1 per order for items
var orders = await _ctx.Orders.ToListAsync();
foreach (var o in orders) { var items = o.Items; } // N lazy load queries!

// ✓ Eager loading - ONE query with JOIN
var orders2 = await _ctx.Orders
    .Include(o => o.Items)
    .ThenInclude(i => i.Product)
    .Include(o => o.Customer)
    .ToListAsync();

// — 3. Project to DTO early - do not load full entities for reads —
// ✗ Loads entire entity graph then maps in C#
var allProducts = await _ctx.Products.Include(p=>p.Category).ToListAsync();
var dtos = allProducts.Select(p => new ProductDto(p.Id, p.Name, p.Category.Name));

// ✓ Project in SQL - only requested columns come back
var dtos2 = await _ctx.Products
    .AsNoTracking()
    .Select(p => new ProductDto(p.Id, p.Name, p.Category.Name)) // SQL: SELECT
    Id,Name,CatName
    .ToListAsync();

// — 4. Always check generated SQL for complex queries —
// In appsettings.json:
// "Microsoft.EntityFrameworkCore": "Information" ← logs all SQL
```

Common EF Core LINQ Patterns

```
// — Conditional filtering (dynamic query building) —
public async Task<List<Product>> SearchAsync(
    string? name, decimal? minPrice, string? category)
{
    var query = _ctx.Products.AsNoTracking().Where(p => !p.IsDeleted);
    if (!string.IsNullOrEmpty(name))
        query = query.Where(p => p.Name.Contains(name));
    if (minPrice.HasValue)
        query = query.Where(p => p.Price >= minPrice.Value);
    if (!string.IsNullOrEmpty(category))
        query = query.Where(p => p.Category == category);
    return await query.OrderBy(p => p.Name).ToListAsync();
}

// — Compiled queries - cache query plan for repeated execution —
private static readonly Func<AppDbContext, int, Task<Product?>> _getById =
    EF.CompileAsyncQuery((AppDbContext ctx, int id) =>
        ctx.Products.AsNoTracking().FirstOrDefault(p => p.Id == id));
```

```
var product = await _getId(_ctx, 42); // faster — skips expression tree parse

// — Raw SQL when LINQ is not enough —
var products2 = await _ctx.Products
    .FromSqlRaw("SELECT * FROM Products WHERE FREETEXT(Name, {0})", searchTerm)
    .AsNoTracking()
    .ToListAsync();
```

EF Core LINQ Do's and Don'ts

✓ DO	✗ DO NOT
Use AsNoTracking() for read-only queries	Track entities you never SaveChanges on
Project with .Select() to DTOs	Load full entities then map in C#
Use .Include() for related data	Access navigation properties in loops (N+1)
Apply .Where() before .ToListAsync()	Call .ToList().Where() (loads all first)
Use CountAsync(), AnyAsync()	Use .Count() > 0 — Any() stops at first match
Use compiled queries for hot paths	Build expressions on every call in hot paths
Check generated SQL (log level Info)	Assume LINQ will generate optimal SQL
Use CancellationToken everywhere	Fire async queries with no cancellation support

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LINQ Performance Tips & Common Mistakes

Any vs Count · Multiple Enumeration · IQueryable Pitfalls

Top LINQ Performance Rules

Rule	Good ✓	Bad ✗	Why
Check existence	<code>list.Any(pred)</code>	<code>list.Count(pred) > 0</code>	<code>Any()</code> stops at first match; <code>Count()</code> scans all
Read-only	<code>AsNoTracking()</code>	default tracking	Skips change tracker = 30-50% faster reads
Avoid multiple enumerations	<code>var list=query.ToList()</code>	Use query twice in <code>foreach/Count</code>	Lazy queries re-execute each time
Filter early	<code>.Where().Select()</code>	<code>.Select().Where()</code>	Filter first reduces projection workload
DB queries	Stay on <code>IQueryable</code>	<code>.AsEnumerable()</code> prematurely	Forces all rows to memory before filtering
Collections	<code>HashSet.Contains()</code>	<code>list.Contains(id)</code> in loop	<code>List.Contains</code> is $O(n)$; <code>HashSet</code> is $O(1)$
String comparison	<code>StringComparer.Ordinal</code>	<code>p.Name.ToLower()==x.ToLower()</code>	Ordinal comparison avoids allocation
Large data	<code>Take(N)</code> + pagination	Load everything, display 10	Memory and bandwidth explosion

Any() vs Count() — The Classic Mistake

```
// ✗ SLOW — Count() iterates the ENTIRE sequence
if (products.Count() > 0) { } // scans ALL elements
if (products.Count(p=>p.IsActive) > 0) { } // same problem

// ✓ FAST — Any() stops at first matching element
if (products.Any()) { } // stops after 1st element found
if (products.Any(p=>p.IsActive)) { } // stops at first active product

// EF Core difference:
// .Count() → SELECT COUNT(*) FROM Products WHERE ...
// .Any() → IF EXISTS (SELECT TOP(1) 1 FROM Products WHERE ...) — faster

// — Multiple enumeration pitfall —
// ✗ query executed TWICE — double the DB round trips
var query = _ctx.Products.Where(p=>p.IsActive);
var count = await query.CountAsync(); // 1st DB query
var list = await query.ToListAsync(); // 2nd DB query

// ✓ Execute once — use the list for count
var list2 = await _ctx.Products.Where(p=>p.IsActive).ToListAsync();
var count2 = list2.Count; // in-memory count, no DB call
```

```
// OR: use a single query that returns both
// (See pagination pattern in Section 11)
```

LINQ Common Mistakes Summary

Mistake	Impact	Fix
IQueryable → AsEnumerable() too early	Full table scan	Keep IQueryable, apply all filters before materialization
Chaining OrderBy().OrderBy()	First sort ignored	Use ThenBy() for secondary sort
Select then Where	Excess processing	Where first to reduce set size
Not calling ToList() — multiple foreach	N executions of query	ToList() once, loop the list
Using List.Contains in LINQ-to-SQL	N+1 or error	Use .Where(p => ids.Contains(p.Id)) — EF generates IN clause
Lazy loading in loops	N+1 queries	Use .Include() for eager loading
Not disposing DbContext	Connection leak	Use scoped DI — AddDbContext() handles disposal
GroupBy without .Select()	No useful data	Always .Select(g => new{g.Key, Items=g.ToList()})

Fundamentals

Q1 What is LINQ and what problem does it solve?

 LINQ Basics

✓ LINQ (Language Integrated Query) is a set of C# features that brings query capabilities directly into the language. It solves the problem of inconsistent APIs for different data sources (arrays use `foreach+if`, databases use SQL strings, XML uses XPath). LINQ provides ONE uniform, type-safe, IntelliSense-powered syntax for all data sources.

```
// One syntax for all sources
var result1 = list.Where(x => x > 5);           // in-memory
var result2 = _ctx.Products.Where(p => p.Price > 5); // SQL database
var result3 = xmlDoc.Descendants("Product")     // XML
    .Where(e => (decimal)e.Element("Price") > 5);
```

 **Tip:** Mention: enabled by lambda expressions, extension methods, anonymous types, expression trees.

Q2 What is the difference between Query Syntax and Method Syntax?

 LINQ Syntax

✓ Query syntax uses SQL-like keywords (`from`, `where`, `select`, `orderby`, `group by`, `join`). Method syntax uses chained method calls (`.Where()`, `.Select()`, `.OrderBy()`). BOTH compile to identical IL — same performance. Method syntax supports ALL operators; query syntax supports a subset. Modern codebases use method syntax 90%+ of the time.

 **Tip:** Remember: query syntax *MUST* start with "from" and end with "select" or "group by". Method syntax can start with any source.

Q3 What is Deferred Execution in LINQ?

 LINQ Execution

✓ Deferred execution means a LINQ query is not executed when it is defined — only when it is iterated or a terminal operator is called. The query is a "recipe" (an `IEnumerable<T>` state machine). Benefits: compose queries incrementally, filter late, no wasted computation. Risks: multiple enumeration re-executes the query.

```
var query = products.Where(p => p.Price > 100); // NOT executed yet
products.Add(new Product{Price=500});          // add after query
var list = query.ToList(); // NOW executes - includes the new product!
```

 **Tip:** Operators triggering immediate execution: `ToList()`, `ToArray()`, `Count()`, `Sum()`, `First()`, `Any()`, `ToDictionary()`

Q4 What is the difference between `IEnumerable<T>` and `IQueryable<T>`?

 LINQ Interfaces

✓ `IEnumerable<T>`: LINQ to Objects — lambdas compile to delegates, execution is in C# memory. `IQueryable<T>`: LINQ to DB — lambdas compile to expression trees, EF Core translates them to SQL. The SAME `.Where()` call on `IQueryable<T>` generates a WHERE clause in SQL, not a C# loop. Never call `AsEnumerable()` on a `DbSet` before filtering — it loads the full table.

```
// IQueryable - SQL WHERE runs on server
_ctx.Products.Where(p => p.Price > 100).ToListAsync()
// -> SELECT * FROM Products WHERE Price > 100
```

```
// IEnumerable - C# filter after full load (bad!)
_ctx.Products.AsEnumerable().Where(p => p.Price > 100).ToList()
// → SELECT * FROM Products then filter 1M rows in C# memory
```

💡 Tip: Key rule: keep the query as *IQueryable* until the last possible moment (*ToListAsync*, *CountAsync*, etc.).

Q5 What is the N+1 problem in LINQ/EF Core?

 EF Core Performance

✅ N+1 occurs when you fetch N items, then lazily load a related entity for each item — resulting in N+1 database queries. Example: fetch 100 orders (1 query), then access *order.Items* per order (100 more queries) = 101 queries. Fix: use *.Include(o => o.Items)* for eager loading — one JOIN query.

```
// ❌ N+1 - 1 + N queries
var orders = await _ctx.Orders.ToListAsync(); // 1 query
foreach(var o in orders) Console.WriteLine(o.Items.Count); // N queries

// ✅ Eager loading - 1 query
var orders2 = await _ctx.Orders.Include(o=>o.Items).ToListAsync();
```

💡 Tip: Mention: enable EF Core SQL logging to detect N+1 (you see the same *SELECT* repeated N times in the output).

Q6 What is the difference between First(), FirstOrDefault(), Single(), and SingleOrDefault()?

 LINQ Element Ops

✅ *First()* = first element, throws if empty. *FirstOrDefault()* = first or null/default, no throw. *Single()* = one element, throws if empty OR more than one. *SingleOrDefault()* = one or null/default, throws if more than one. Use *Single()* for lookups where uniqueness is required (e.g., by primary key); use *First()* when multiple results are acceptable.

💡 Tip: *Single()* scans the WHOLE sequence to ensure no second match — *O(n)*. *First()* stops at the first match — *O(1)* best case. In EF Core, both generate efficient SQL.

Q7 What is SelectMany and when do you use it?

 LINQ Projection

✅ *SelectMany* projects each element to a sequence, then FLATTENS all resulting sequences into one. Use it when each element contains a collection and you want a flat list of all nested items. It replaces nested *foreach* loops. Example: from a list of *Orders*, get all *OrderItems* flat.

```
var allItems = orders.SelectMany(o => o.Items);
// Instead of: foreach(var o in orders) foreach(var i in o.Items) list.Add(i)

var words = sentences.SelectMany(s => s.Split(" "));
```

💡 Tip: *SelectMany* is *Select* + *Flatten*. If *Select* returns *IEnumerable<IEnumerable<T>>*, *SelectMany* gives *IEnumerable<T>*.

Q8 How does GroupBy work? What does it return?

 LINQ Grouping

✅ *GroupBy* returns *IEnumerable<IGrouping<TKey,TElement>>*. Each *IGrouping* has a *.Key* (the group key) and implements *IEnumerable<TElement>* (the items in the group). *GroupBy* is DEFERRED — it does not execute until iterated. Always follow *GroupBy* with *.Select()* to project the groups into useful shapes.

```
var groups = products.GroupBy(p => p.Category)
    .Select(g => new {
        Category = g.Key,
        Count     = g.Count(),
    });
```

```
        Total    = g.Sum(p => p.Price)
    });
```

💡 **Tip:** *GroupBy vs ToLookup: GroupBy is lazy (re-evaluates each iteration). ToLookup is eager — built once, accessed many times like a dictionary.*

Q9 What is Aggregate() and give a real example?

🔗 LINQ Aggregation

✅ **Aggregate()** is a general-purpose fold/reduce operator. It applies an accumulator function to each element, carrying forward a running value. It replaces manual foreach+temp-variable patterns. Three overloads: no seed (uses first element), with seed, with seed+result selector.

```
// Without seed - uses first element as initial accumulator
var product = new[]{1,2,3,4,5}.Aggregate((acc,n) => acc * n); // 120

// With seed
var sum = new[]{1,2,3}.Aggregate(100, (acc,n) => acc+n); // 106

// Build CSV
var csv = products.Aggregate("",
    (acc,p) => acc + (acc.Length>0?",";"") + p.Name);
```

💡 **Tip:** *Aggregate is the Swiss army knife of aggregation. Sum/Count/Min/Max are all special cases of Aggregate with built-in optimisation.*

Q10 What is PLINQ and when should you use it?

🔗 LINQ Advanced

✅ **PLINQ** (Parallel LINQ) processes a sequence in parallel across multiple CPU cores using `.AsParallel()`. Good for CPU-bound processing on large in-memory datasets. Do NOT use for I/O-bound work (use `async/await` instead) or small collections (thread overhead exceeds benefit). Use `.AsOrdered()` if result order must match input order.

```
// PLINQ - parallel in-memory processing
var results = largeNumbers
    .AsParallel()
    .WithDegreeOfParallelism(4)
    .Where(n => IsPrime(n))
    .ToList();

// Preserve order (slight overhead)
var ordered = data.AsParallel().AsOrdered().Select(Transform).ToList();
```

💡 **Tip:** *PLINQ is for CPU-bound in-memory work. Never use for database queries or I/O. Always benchmark — PLINQ has overhead that makes it slower for small collections.*

Advanced Questions

Q11 Can you write a custom LINQ extension method?

 LINQ Extension Methods

✓ Yes — LINQ operators are just extension methods on `IEnumerable<T>`. You can write your own following the same pattern: a static class, a static method, first parameter "this `IEnumerable<T>` source", return `IEnumerable<T>` for deferred execution (use `yield return`).

```
public static class LinqExtensions {
    // Custom operator: take while sum does not exceed max
    public static IEnumerable<T> TakeUntilSum<T>(
        this IEnumerable<T> source, Func<T, decimal> selector, decimal maxSum) {
        decimal running = 0;
        foreach (var item in source) {
            running += selector(item);
            if (running > maxSum) yield break;
            yield return item;
        }
    }
}

// Usage
var affordable = products.OrderBy(p=>p.Price).TakeUntilSum(p=>p.Price, budget);
```

 **Tip:** Custom operators should return `IEnumerable<T>` with `yield return` for deferred execution. This is how all built-in LINQ operators work.

Q12 What is the difference between Any() and All()?

 LINQ Quantifiers

✓ `Any()` returns true if AT LEAST ONE element satisfies the condition (or if sequence is non-empty when called with no predicate). `All()` returns true if EVERY element satisfies the condition — and returns TRUE for an empty sequence (vacuous truth). Both are short-circuit: `Any()` stops at first match, `All()` stops at first non-match.

```
var hasExpensive = products.Any(p => p.Price > 500); // true if at least 1
var allActive    = products.All(p => p.IsActive);    // true if ALL are active
var isEmpty      = !products.Any();                 // no elements

// Vacuous truth — All() on empty = true
new List<Product>().All(p => p.Price < 0) // TRUE — empty list satisfies Any condition
// This catches developers off guard!
```

 **Tip:** `All()` on an empty sequence returns TRUE (logical vacuous truth). Always check `Any()` first if you need non-empty validation.

Q13 What is Expression<Func<T>> vs Func<T>?

 LINQ Expression Trees

✓ `Func<T, bool>` is a compiled delegate — executes C# code. `Expression<Func<T, bool>>` is an expression tree — a data structure representing the code that can be analysed, modified, and translated (e.g., to SQL by EF Core). `IQueryable.Where()` takes `Expression<Func<T, bool>>`; passing a `Func` causes all rows to load first.

```
// Func — compiled, runs in C#
Func<Product, bool> func = p => p.Price > 100;
_ctx.Products.Where(func).ToList(); // loads ALL products first! (IEnumerable path)

// Expression<Func> — expression tree, translates to SQL
Expression<Func<Product, bool>> expr = p => p.Price > 100;
_ctx.Products.Where(expr).ToList(); // WHERE Price > 100 in SQL
```

 **Tip:** The EF Core `DbSet.Where()` signature takes `Expression<Func<T, bool>>`. If you accidentally pass a `Func`, the compiler lets you — but EF Core silently switches to client evaluation.

Q14 How do you build a dynamic LINQ query?

 LINQ Advanced

✅ Build IQueryable chains conditionally by adding .Where() clauses only when needed. Since IQueryable is lazy, you can incrementally build the query before executing it. This is the correct way to implement search/filter APIs without concatenating SQL strings.

```
public IQueryable<Product> BuildQuery(SearchParams p) {
    var q = _ctx.Products.AsNoTracking().Where(x => !x.IsDeleted);
    if (!string.IsNullOrEmpty(p.Name)) q = q.Where(x => x.Name.Contains(p.Name));
    if (p.MinPrice.HasValue) q = q.Where(x => x.Price >=
p.MinPrice.Value);
    if (p.MaxPrice.HasValue) q = q.Where(x => x.Price <=
p.MaxPrice.Value);
    if (p.CategoryId.HasValue) q = q.Where(x => x.CategoryId ==
p.CategoryId);
    return q;
}
```

 **Tip:** This is the real-world answer to "how do you implement search?" in interviews. The query is built programmatically — no string concatenation, full type safety.

Q15 What are compiled queries in EF Core?

 EF Core Advanced

✅ EF Core normally parses the expression tree and generates SQL on every query call. Compiled queries cache this translation after the first call — subsequent calls skip the expensive expression parsing. Use for frequently called queries in performance-critical paths.

```
// Define once at class level
private static readonly Func<AppDbContext, int, Task<Product?>> GetByIdCompiled =
    EF.CompileAsyncQuery((AppDbContext ctx, int id) =>
        ctx.Products.AsNoTracking().FirstOrDefault(p => p.Id == id));

// Use in method — much faster on repeat calls
var product = await GetByIdCompiled( ctx, productId);
```

 **Tip:** Compiled queries are 20-40% faster for frequently executed queries because they skip the LINQ expression tree compilation step.

All LINQ Operators at a Glance

Category	Operator	Returns	Deferred?	Quick Note
Filtering	Where(pred)	IEnumerable<T>	✓ Yes	Most used. Supports index overload.
Filtering	OfType<T>()	IEnumerable<T>	✓ Yes	Type-safe filter on object sequences.
Projection	Select(selector)	IEnumerable<R>	✓ Yes	One-to-one transform.
Projection	SelectMany(selector)	IEnumerable<R>	✓ Yes	Flatten nested collections.
Sorting	OrderBy(key)	IOrderedEnumerable	✓ Yes	Always use ThenBy for secondary sort.
Sorting	OrderByDescending(key)	IOrderedEnumerable	✓ Yes	Descending primary sort.
Sorting	ThenBy(key)	IOrderedEnumerable	✓ Yes	Secondary sort (chain after OrderBy).
Sorting	ThenByDescending(key)	IOrderedEnumerable	✓ Yes	Secondary descending sort.
Sorting	Reverse()	IEnumerable<T>	✓ Yes	Reverses sequence (not sort).
Grouping	GroupBy(key)	IEnumerable<IGrouping>	✓ Yes	Lazy. Each group has .Key.
Grouping	ToLookup(key)	ILookup<K,V>	✗ No	Eager. Missing key → empty.
Joining	Join(inner,ok,ik,result)	IEnumerable<R>	✓ Yes	Inner join (both must match).
Joining	GroupJoin(inner,ok,ik,result)	IEnumerable<R>	✓ Yes	Left outer join pattern.
Joining	Zip(second, result)	IEnumerable<R>	✓ Yes	Element-by-element pairing.
Aggregation	Count() / Count(pred)	int	✗ No	Use Any() for existence check.

Category	Operator	Returns	Deferred?	Quick Note
Aggregation	LongCount()	long	✗ No	For > int.MaxValue elements.
Aggregation	Sum(selector)	numeric	✗ No	Returns 0 if empty.
Aggregation	Min(selector)	T	✗ No	Throws if empty. Use MinBy (.NET6).
Aggregation	Max(selector)	T	✗ No	Throws if empty. Use MaxBy (.NET6).
Aggregation	Average(selector)	double	✗ No	Throws if empty.
Aggregation	Aggregate(func)	T	✗ No	General fold/reduce.
Set	Distinct()	IEnumerable<T>	✓ Yes	Remove duplicates.
Set	DistinctBy(key) .NET6	IEnumerable<T>	✓ Yes	Deduplicate by key selector.
Set	Union(second)	IEnumerable<T>	✓ Yes	All unique from both.
Set	Intersect(second)	IEnumerable<T>	✓ Yes	Common elements only.
Set	Except(second)	IEnumerable<T>	✓ Yes	In first but not second.
Partitioning	Take(n)	IEnumerable<T>	✓ Yes	First n elements.
Partitioning	Skip(n)	IEnumerable<T>	✓ Yes	Skip first n elements.
Partitioning	TakeWhile(pred)	IEnumerable<T>	✓ Yes	Take until pred is false.
Partitioning	SkipWhile(pred)	IEnumerable<T>	✓ Yes	Skip until pred is false.
Partitioning	Chunk(size) .NET6	IEnumerable<T[]>	✓ Yes	Split into batches.
Element	First() / First(pred)	T	✗ No	Throws if empty/no match.
Element	FirstOrDefault()	T?	✗ No	Null/default if not found.
Element	Single() / Single(pred)	T	✗ No	Throws if 0 or >1 result.

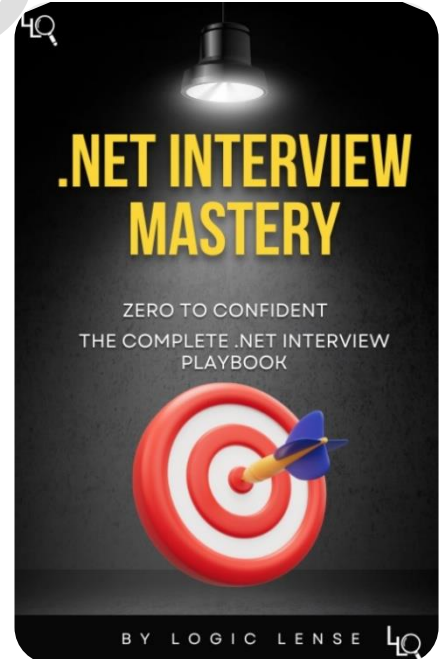
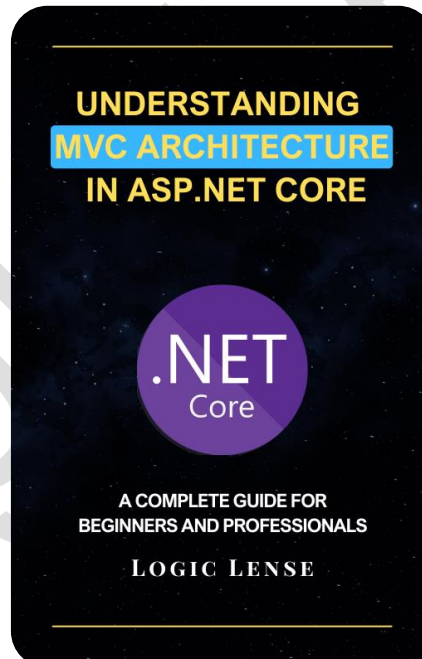
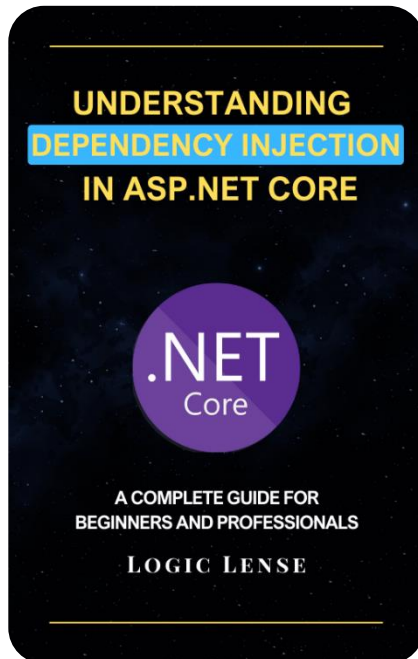
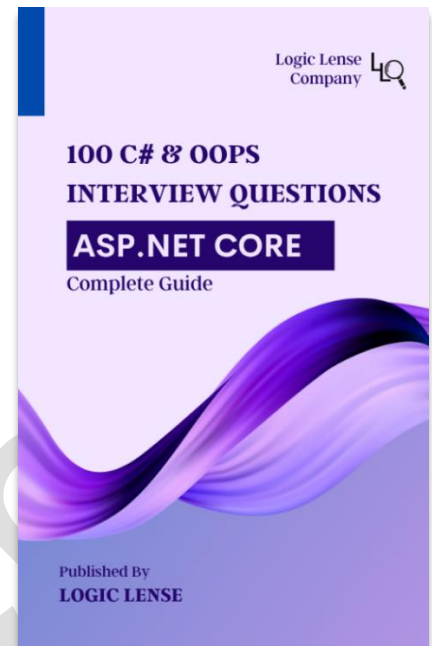
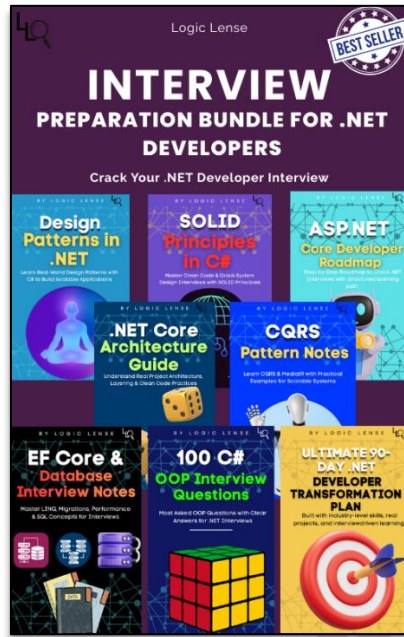
Category	Operator	Returns	Deferred?	Quick Note
Element	SingleOrDefault()	T?	✗ No	Null/default if 0; throws if >1.
Element	Last() / LastOrDefault()	T / T?	✗ No	Last element or default.
Element	ElementAt(index)	T	✗ No	By index; throws if out of range.
Element	ElementAtOrDefault(index)	T?	✗ No	By index or default.
Element	DefaultIfEmpty()	IEnumerable<T>	✓ Yes	Returns [default] if empty.
Quantifier	Any() / Any(pred)	bool	✗ No	Short-circuits at first match.
Quantifier	All(pred)	bool	✗ No	True for empty sequence (vacuous).
Quantifier	Contains(value)	bool	✗ No	Linear search. Use HashSet for O(1).
Conversion	ToList()	List<T>	✗ No	Forces execution, snapshot.
Conversion	ToArray()	T[]	✗ No	Forces execution, fixed size.
Conversion	ToDictionary(key)	Dictionary<K,V>	✗ No	Throws on duplicate keys.
Conversion	ToLookup(key)	ILookup<K,V>	✗ No	Multiple values per key.
Conversion	AsEnumerable()	IEnumerable<T>	✓ Yes	Switch from IQueryable to in-memory.
Conversion	AsQueryable()	IQueryable<T>	✓ Yes	Wrap IEnumerable as IQueryable.
Generation	Enumerable.Range(start,count)	IEnumerable<int>	✓ Yes	Generate integer range.
Generation	Enumerable.Repeat(elem,count)	IEnumerable<T>	✓ Yes	Generate repeated elements.
Generation	Enumerable.Empty<T>()	IEnumerable<T>	✓ Yes	Empty sequence.
Parallel	AsParallel()	ParallelQuery<T>	✓ Yes	CPU-bound parallel processing.

Category	Operator	Returns	Deferred?	Quick Note
Parallel	AsOrdered()	ParallelQuery<T>	☒ Yes	Preserve order in PLINQ.

☆ Last-Minute Revision — One-Liners

Concept	One-Line Answer
LINQ	Language Integrated Query — uniform query syntax for any data source
Deferred Execution	Query executes only when iterated — ToList()/foreach triggers it
IEnumerable vs IQueryable	IEnumerable=C# code; IQueryable=expression tree translated to SQL
N+1 Problem	1 query + N per-row queries — fix with .Include() eager loading
Select vs SelectMany	Select=transform 1:1; SelectMany=flatten nested collections
GroupBy	Returns IGrouping<K,V> — deferred. Use .Select() to project groups.
ToLookup	Eager GroupBy — builds lookup table once, safe missing-key access
Any() vs Count(>0)	Any() is O(1) best-case — Count() always iterates everything
Single vs First	Single=exactly one expected; First=first of possibly many
AsNoTracking()	Skip EF Core change tracker for 30-50% faster read queries
Expression Tree	Code-as-data structure — how EF Core translates LINQ to SQL
yield return	Makes a method a lazy iterator — used by all deferred operators
PLINQ	AsParallel() for CPU-bound in-memory data — not for I/O or DB
Compiled Query	EF.CompileAsyncQuery() — cache SQL generation for hot paths

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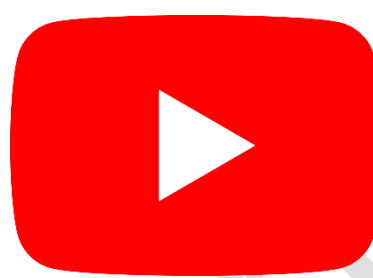


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